

How AI is Affecting Farmed Aquatic Animals

Part 2: Deployment

Animal Welfare Department

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AI in Aquaculture Research at Rethink Priorities

Rethink Priorities' *How AI is Affecting Farmed Aquatic Animals* is a three-part series that examines AI technologies being developed to address challenges in the aquaculture industry. As global aquaculture faces economic and technical constraints, this series examines whether and how the industry is turning to AI solutions to facilitate expansion. The series evaluates the [innovation](#), deployment, and animal welfare impacts of these technologies to help animal advocates and funders identify which developments should be endorsed and what actions should be taken to prevent increased animal suffering.

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AI in Farmed Aquaculture Series

Artificial intelligence (AI) introduces new capabilities to animal agriculture that could alter production methods, economic structures, and animal welfare outcomes. Responding strategically requires an understanding of how quickly such changes will unfold, whether they will benefit or harm animal welfare, and what interventions will remain relevant. In this three-part series, we take a close look at how AI will be used over the next five years in aquaculture, which collectively farms hundreds of billions of animals each year for food.

Part 2: Deployment

This report analyzes the current state of AI deployment in aquaculture, examining where AI-aquaculture tools are being used in practice, the species and regions with the greatest AI presence, what is driving and constraining adoption, and how deployment is expected to evolve. The final report will address *Welfare Effects*.

Key findings from part two of this series:

- We found evidence of deployment across 71 countries for 66 aquaculture-specialist companies, concentrated in Europe, Latin America, and Southeast Asia.
- Norway and Scotland are the largest exporter countries of aquaculture AI in our dataset.
- Salmon and shrimp have the broadest deployment footprints—present in more countries than any other species in our dataset.
- Experts estimate that ~15% of all salmon producers and <10% of all shrimp producers currently use AI tools, rising to ~75% and ~25% among top producers.
- The primary barrier to adoption is reported to be an unclear return on investment.

Executive Summary

This report analyzes the current state of AI deployment in aquaculture. In the previous report, we identified which AI aquaculture products currently exist. In this report, we examine where those products are being used and to what extent their deployment may change in the short term (roughly the next five years). To do this, we assessed deployment evidence for 66 aquaculture-specialist companies through targeted web searches, supplemented by three expert interviews and a review of 16 academic papers on barriers to adoption.

- From the 91 companies identified in [Part 1](#), we focused on those specializing in aquaculture and found deployment evidence for 66 companies, spanning 71 countries and 276 company-country pairs.
- **Where:** Deployment is concentrated in Europe, Latin America, and Southeast Asia, with Norway and Scotland the largest net exporters of aquaculture AI.
 - In our dataset, Chile has more foreign-headquartered companies deploying there than any other single country and has no domestically-headquartered AI companies.
 - Southeast Asia is the most self-sufficient region, with the majority of the companies we found deploying there also headquartered within the region.
- **Which species:** Salmon and shrimp AI tools are deployed across more countries than for any other species in our dataset, consistent with our finding that they are the most targeted species in [Part 1](#).
 - Salmon deployment is more geographically concentrated, particularly in Norway and Chile; shrimp deployment is more evenly distributed.
- **Who:** Experts estimate that top producers have much higher rates of adopting at least one AI tool than the average producer: ~75% compared to ~15% of salmon farms, and ~25% compared to ~10% of shrimp producers.
 - This gap likely reflects the high cost of current AI systems and the difficulty of justifying that cost at smaller scales of production.
- **Current barriers:** Experts report that the primary barrier to adoption is an unclear return on investment and that tools with demonstrable returns, such as biomass estimation in salmon and feed optimization in shrimp, have gained the broadest traction.

- **Outlook:** AI adoption is expected to expand rapidly in the short term, with one expert projecting deployment across more than 70% of farmed shrimp hectares in Ecuador within five years.

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Background

This report is the second in a three-part series examining how AI is affecting farmed aquatic animals. In [Part 1](#) of this series, we analysed the current state of AI innovation in aquaculture and found 91 companies with AI-enabled technologies that have direct implications for farmed animal welfare. You can see the database [here](#). We found that innovation is concentrated primarily in the US and Norway, predominantly targets high-value species such as salmon and shrimp, and most commonly addresses stock and growth management. Part 2 builds on this foundation by examining where and to what extent those tools are being deployed.

Scope

This report draws on three sources of evidence: a desk-research investigation into the deployment of AI tools from the list of companies we identified in [Part 1](#), three expert interviews, and a Claude-assisted review of academic literature on barriers to AI adoption.

For the deployment analysis, we took the 91 companies identified in [Part 1](#) and assessed where each had evidence of deploying their products (see our [database](#)). We excluded companies that were not primarily focused on aquaculture, and for three aquaculture-only companies, we could not find any deployment information, leaving 66 companies for the analysis. For each company, we used Claude to conduct targeted web searches combining company names with deployment-related search terms (including "deployment," "locations," "farms," and "pilots"), and asked Claude to return results organized by country with links. We then checked all the links manually. We recorded a deployment location only where a company explicitly stated it was operating or had deployed technology in a specific country, as found on company websites, press releases, third-party articles, or other online resources. We included deployments at all stages—pilot, field trial, and commercial-scale—and recorded them at the company level rather than the product level. Countries mentioned only as target markets, future expansion plans, or office locations without corroborating evidence of on-site product use were excluded. Full inclusion criteria are provided in the [Appendix](#).

To supplement the desk research, **we conducted interviews with three experts with direct experience of AI deployment in aquaculture**. Expert backgrounds are described in the [Appendix](#). The interviews covered current deployment patterns by species and region, barriers to adoption, and expectations for near-term growth. Given that two of the three experts specialize in salmonids and one in shrimp, findings from the expert interviews are most reliable for these species and should be interpreted with caution when generalized to others.

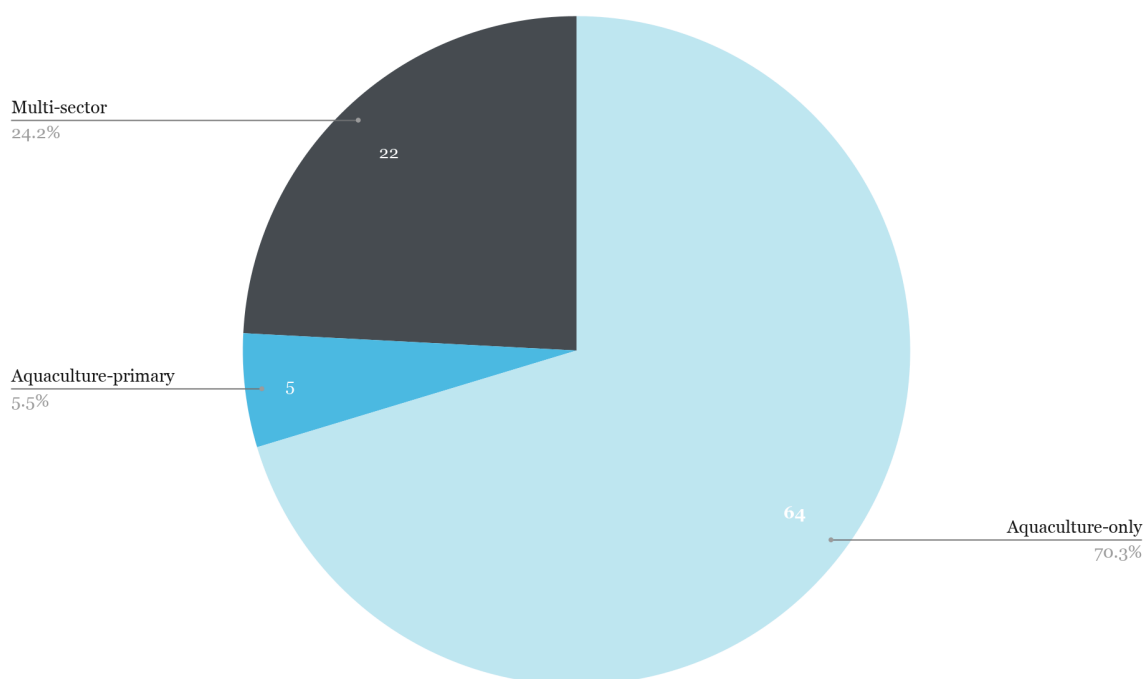
To examine barriers to AI adoption, **we conducted a timeboxed three-hour search for academic literature** and retained 16 papers for which the full text was available—13 focused primarily on aquaculture and three drew on broader agricultural contexts. We used Claude to extract barrier themes from each paper and compared these against expert views, selecting themes with the strongest consensus across both sources. Further details on this process are provided in the [Appendix](#).

This report does not constitute a comprehensive review of AI deployment in aquaculture. Our desk research identifies deployment evidence that is publicly available in English-language sources, which likely understates deployment activity in regions with limited online presence or where companies publish primarily in other languages. Expert interview coverage is heavily weighted towards salmon and shrimp, and towards Europe and Latin America. See other [ways we could be wrong](#) below.

Results

Three-Quarters of Companies in Our Database Specialize in Aquaculture

Figure 1: Distribution of 91 companies making AI-enabled tools for aquaculture, broken down by degree of company specialization.



Of the 91 companies we found in [Part 1](#) that produce AI-enabled tools for aquaculture, we found that over three-quarters of companies specialized in products solely for aquaculture (~70%), or aquaculture with minor applications in a closely related field (~6%) (see the [Appendix](#) for definitions). For three companies in the aquaculture-only category, we could not find any deployment information, leaving 66 companies for the deployment analysis that follows.

Europe Leads Company Presence, Followed by Latin America and Southeast Asia

To map where these 66 companies deploy their products, we recorded every country in which each company had evidence of deployment. Our inclusion criteria for deployment locations are defined in the [Appendix](#). Among the 66 companies specializing in AI-aquaculture technology, we found 276 company-country pairs in total.¹

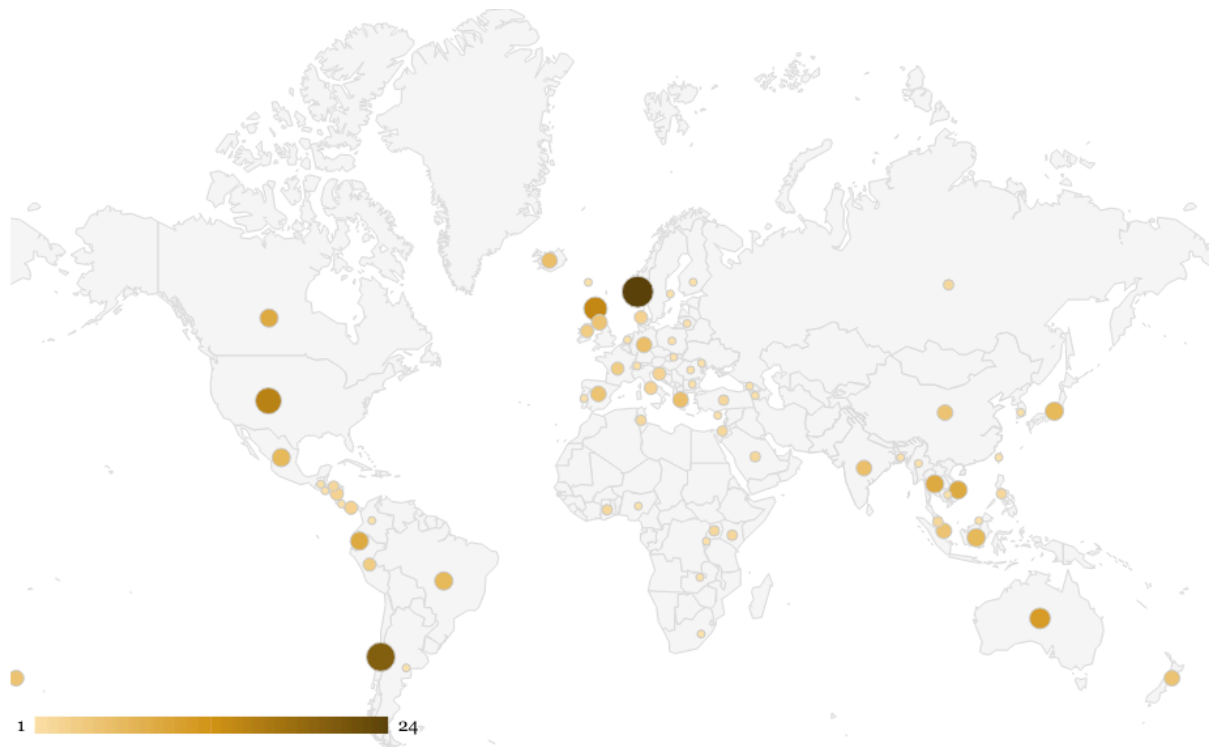
Box 1: Company-Country Pairs

A company-country pair is a unique combination of one company and one country. A company deploying to three countries contributes three pairs, regardless of how many species or farms it targets in each.

The distribution of company-country pairs is shown in Figure 2.

¹ The total counts company-country pairs per headquarters location, so a company with dual headquarters counts each separately: e.g. a company headquartered in both Norway and the US deploying only to Norway contributes two pairs: one for the Norway deploying to Norway, and one for the US deploying to Norway. Only 5/66 companies analyzed have dual headquarters.

Figure 2: Geographic distribution of AI-aquaculture technology deployed by 66 companies, counting each company-country pair once regardless of species targeted. Represents deployment to 71 unique countries from companies with headquarters in 31 unique countries.



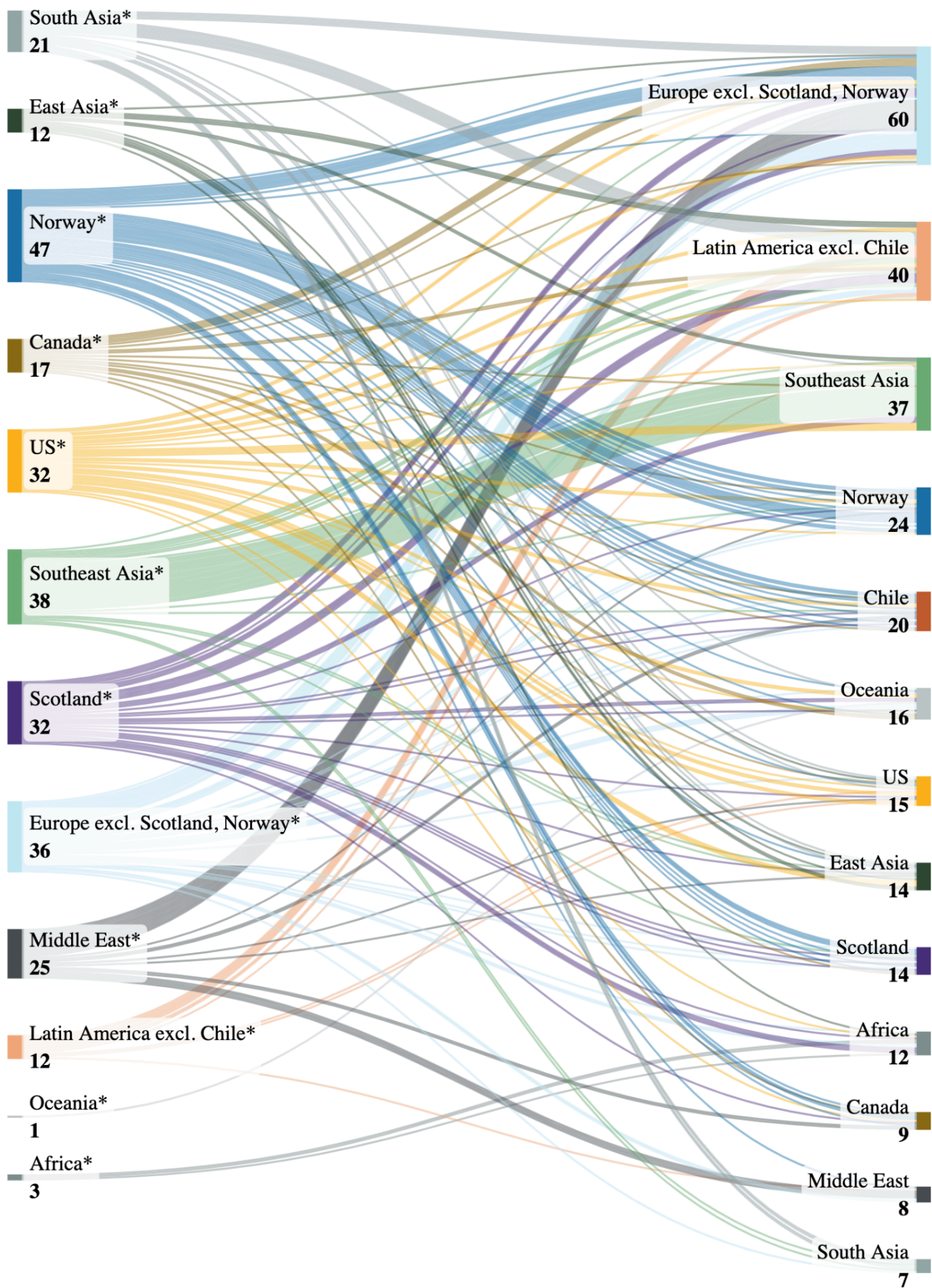
The most frequently represented regions were Europe, Latin America, and Southeast Asia, with 98, 60, and 37 company-country pairs across 26, 13, and 9 deployment countries, respectively. **Within Europe, Norway and Scotland accounted for the largest shares**, with 24 and 14 company-country pairs, respectively. **Within Latin America, Chile dominated** with 20 company-country pairs present. Outside these three regions, the US and Australia recorded notable presences with 15 and 11 company-country pairs, respectively.

Aquaculture AI Flows Predominantly From Northern Europe into Latin America and the Rest of Europe, With Innovation in Southeast Asia Largely Staying Within Southeast Asia

Figure 3 maps the headquarters locations of 66 aquaculture-focused AI companies against their deployment countries. This illustrates that AI aquaculture tools have a larger global reach than only the countries in which they are produced, and that **the welfare implications of a tool developed in one country could extend to animals in many others**. For advocates

seeking to influence the design of AI tools deployed in their region, knowing where the developing companies are headquartered could inform where to direct that engagement.

*Figure 3: Sankey diagram, created on [SankeyMATIC](#), showing the headquarters location (left) of 66 aquaculture-focused AI companies and where their products are deployed (right), independent of species. The width of each flow is proportional to the number of company-country pairs. For example, a company deploying AI products to three countries counts as three pairs. There are a total of 265 company-country pairs across the 66 companies, where five of the included companies hold dual headquarters, resulting in 276 headquarters-to-deployment pairs mapped on the diagram. Numbers on the left reflect the total company-country pairs for companies headquartered in that region; numbers on the right reflect the total company-country pairs for deployments received by that region. *denotes location of the company headquarters.*



Norway and Scotland Are the Two Largest Net Exporters of Aquaculture AI by Absolute Net Flow.

Norway and Scotland are the largest net exporters by absolute net flow. Norway is the headquarters of 14 companies in our dataset. These companies generate 47 company-country pairs in total, of which 13 are within Norway itself, and 33 are abroad. Scotland is the headquarters of three companies in our dataset. These companies generate 32 company-country pairs in total, of which three are within Scotland itself, and 29 are abroad.

Europe (excluding Scotland and Norway) and Latin America (excluding Chile) are the largest net importers by absolute net flow. Europe (excluding Scotland and Norway) has 60 company-country pairs, of which 12 involve companies headquartered within the region, and 48 involve companies headquartered elsewhere. Latin America (excluding Chile) has 40 company-country pairs, of which eight involve companies headquartered within the region, and 32 involve companies headquartered elsewhere. **This picture is supported by experts, who reported AI adoption to be most advanced in Europe and Latin America.**

Chile has the highest number of foreign-headquartered companies deploying there in our dataset, and no domestically-headquartered companies. Experts corroborated that **Chile currently imports most of its AI systems**, adding these imports are primarily from Norwegian companies. In our own dataset, eight of the 20 companies present in Chile are headquartered in Norway—more than from any other region or single country (40%) deploying to Chile.

Southeast Asia is the most self-sufficient region, with 24 of the 38 company-country pairs generated by Southeast Asian-headquartered companies deployed within the region itself.

Salmon and Shrimp Have the Largest AI Deployment Footprints

To assess how deployment varies by species, we count the number of companies deploying AI-aquaculture tools in each country for each species separately.

Box 2: Deployment Instances

A deployment instance is a unique combination of one company, one country, and one species. A single company operating in 10 countries, and 10 companies each operating in one country, would both contribute 10 deployment instances for a given species. A high deployment instance count for a given species may therefore reflect many

companies targeting that species, a wide geographical spread of products targeting that species, or both.

Deployment instances are comparable across species and countries, but their sum across both represents only total deployment activity across the dataset and cannot be decomposed into a count of companies, countries, or species.

Shrimp and salmon have the broadest deployment footprint in the dataset in absolute terms (136 and 131 instances, respectively), followed by trout (88), sea bass (69), sea bream (70), tilapia (56), and yellowtail (56).²

Table 1 shows the top targeted species, as defined by deployment instances, for the five countries with the highest number of company-country pairs (as shown in Figure 2).

Table 1: Top-ranked targeted species by deployment instances for the five countries with the highest number of company-country pairs. Where species are tied, all are shown.

	Top targeted species (by number of deployment instances)		
Country	1st	2nd	3rd
Norway	Salmon (20)	Trout (9)	Sea bass/sea bream (7)
Chile	Salmon (17)	Trout (8)	Sea bass/sea bream (7)
US	Shrimp/salmon (8)	Trout/yellowtail (6)	Sea bass/sea bream (4)
Scotland	Salmon (12)	Trout (8)	Shrimp/charr/sea bass/sea bream/tilapia (3)
Australia	Salmon (5)	Shrimp (4)	Sea bass/sea bream/tilapia (3)

In [Part 1](#) of this series, we found that salmon and shrimp co-ranked as the top-most targeted species by AI-aquaculture product innovation. We find that the picture for salmon is reflected in deployment: salmon has the highest overall AI-aquaculture presence and dominates the top spot for most targeted species in every country. For shrimp, the numbers suggest differently: despite an equally large total deployment footprint, shrimp reaches top rank only

² This is consistent with our finding at the innovation stage that these species had the most companies targeting them. However, the two findings are not independent: species with more dedicated AI innovation would be expected to show higher deployment instances simply as a result. The deployment ranking across species should not therefore be read as evidence that any species has seen disproportionately high or low uptake of the tools developed for it.

in the US, where it ties with salmon. Figure 4 and Figure 5, discussed below, show the full country-level distribution for these two species.³

The following sections list the countries with the highest salmon and shrimp deployment instances in our dataset.

Salmon

- Norway (20 deployment instances)
- Chile (17 deployment instances)
- Scotland (12 deployment instances)⁴
- Canada and the United States (8 deployment instances each)

These five countries account for ~50% of AI-aquaculture tools deployed targeting salmon (65/131 deployment instances across 44 countries).

Consistent with our data, experts noted that highest adoption rates are seen in Chile, Norway, and Scotland, though this is in part expected given that countries with the largest Atlantic salmon farming industries could reasonably be anticipated to show the highest absolute deployment regardless of adoption rates.

The experts reported that Norway leads in AI testing, attributing this to the ability of salmon farming companies to absorb the cost premium of current AI systems. Conversely, they suggested that Chile leads in adoption due to a younger ([UNCTAD, 2006](#)), more hierarchical management structure than Norway ([Dessler, 2017](#), p. 152) that enables faster top-down adoption decisions and rapid scale-up once a technology is trialled.

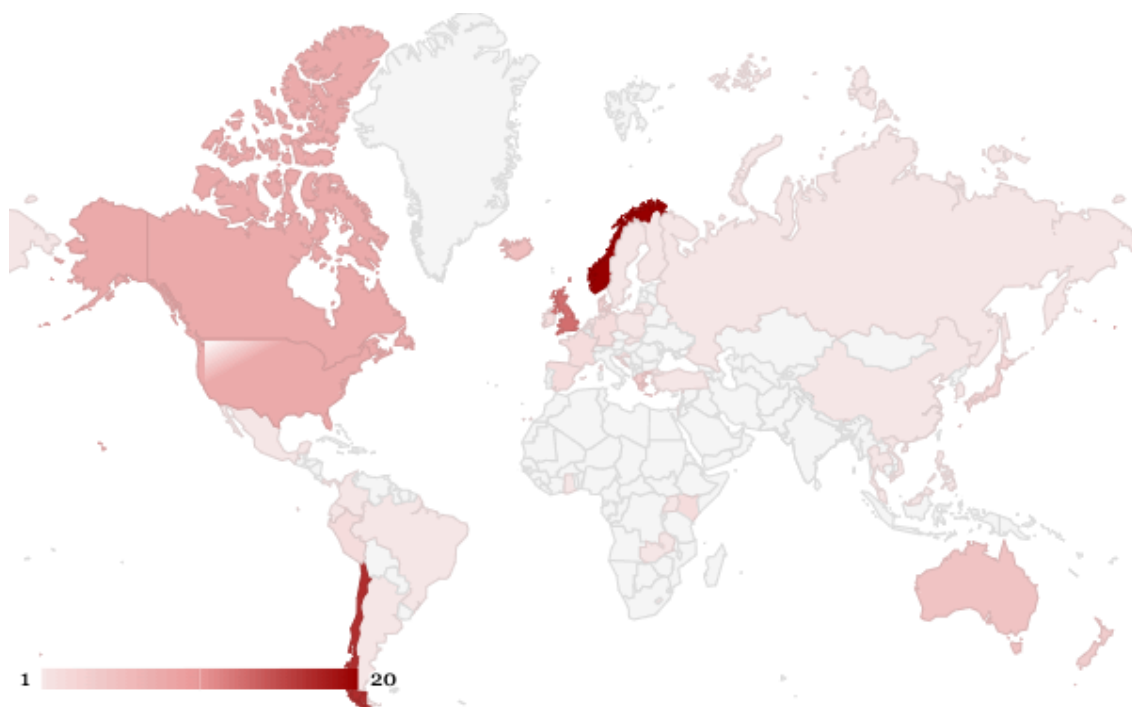
Experts estimate that AI technology is used by approximately 15% of all salmon producers, rising to upwards of 75% when looking at only top salmon producers.⁵

Figure 4: Geographic distribution of AI-aquaculture deployment instances targeting salmon across 44 countries, drawn from a dataset of 66 companies. Graph shows 131 deployment instances, where 1 count represents one company present in one country for that species.

³ For Figure 4 and Figure 5, Scotland and the United Kingdom are merged into a single entry (United Kingdom), as are the Faroe Islands and Denmark, because Google Sheets Geocharts cannot display territories separately from their parent countries. Where a company listed Scotland, the UK, or both, this was counted as a single deployment instance.

⁴ We found one further salmon deployment instance in the UK where Scotland was not specified.

⁵ The term “top producers” was not formally defined in the interviews, with experts variously referring to “top 10 producers” and “very large players.” We interpret this as referring to the largest aquaculture companies by production volume, though the precise definition was not clarified.



Shrimp

- Ecuador, Vietnam, Thailand, and the United States (8 deployment instances each)
- Mexico, Indonesia, and the United Kingdom (6 deployment instances each)

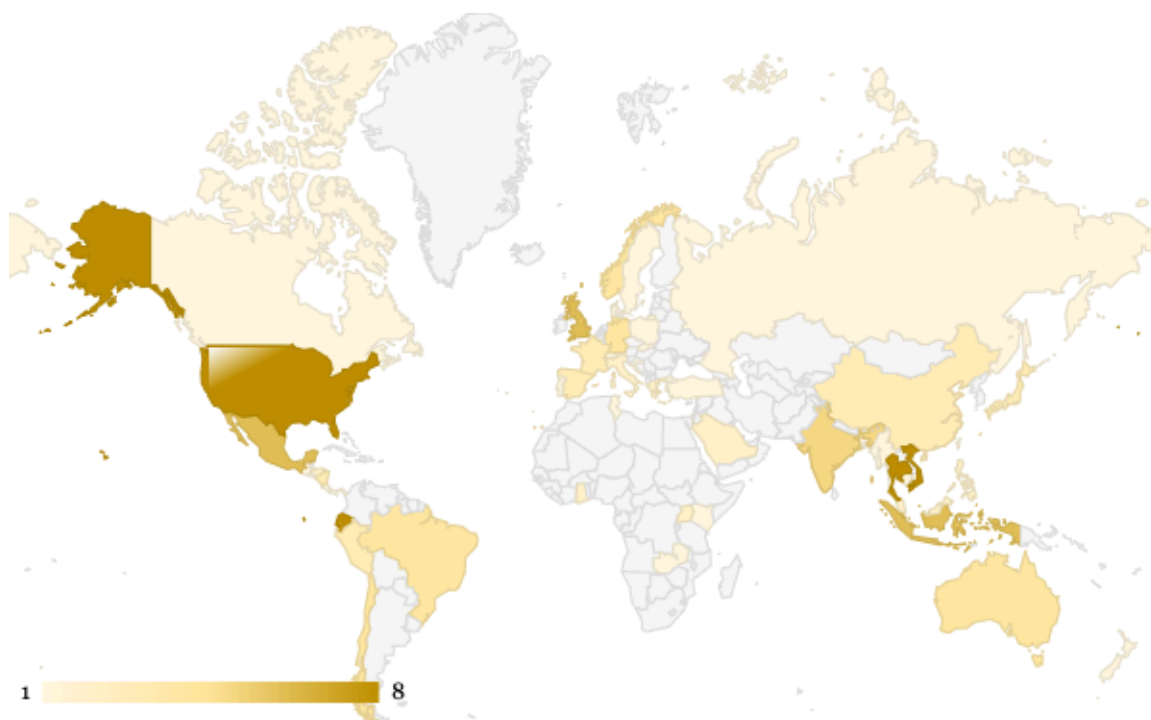
These 7 countries account for ~37% of AI-aquaculture tools deployed targeting shrimp (50/136 deployment instances across 52 countries).

Consistent with our findings, experts ranked Ecuador first by some distance. One expert further specified Mexico, Brazil, and Vietnam as other developing markets. Notably, despite the US ranking joint first with Ecuador for shrimp deployment instances in our dataset, it was not identified by any expert as a significant shrimp AI-adoption market. This discrepancy may simply reflect the scale of US shrimp production: one expert noted that US shrimp production is negligible, suggesting that even if adoption rates of AI technology among US producers are relatively high, it may not translate into the US being considered a significant AI-adoption market in global terms.

Experts estimate that AI technology is used by approximately 5–10% of all shrimp producers and up to 20–30% among top shrimp producers.⁶

Figure 5: Geographic distribution of AI-aquaculture deployment instances targeting shrimp across 52 countries, drawn from a dataset of 66 companies. Graph shows 136 deployment instances, where 1 count represents one company present in one country for that species.

⁶ See footnote above.



Of the 66 companies included on the graphs, 27 target salmon and 26 target shrimp, meaning the average salmon company is deployed in ~4.9 countries, and the average shrimp company in ~5.2 countries. However, the average deployment instances per company do not capture the evenness of the spread across countries. To quantify this geographic concentration, we calculated the Gini coefficient for salmon and shrimp deployment instances. The Gini coefficient is a measure of concentration ranging from 0 to 1, here applied to the geographic distribution of deployment instances across countries. A value of 0 indicates perfectly equal distribution across all countries where companies operate; a value of 1 indicates complete concentration in a single country. Salmon has a Gini coefficient of 0.54, compared to 0.41 for shrimp, indicating that **salmon AI-tool deployment is more geographically concentrated in a smaller number of markets**—particularly Norway and Chile. Shrimp deployment is less geographically concentrated than salmon's, with company activity distributed more evenly across the countries in which they operate.

Experts Suggest Biomass Tools Lead On-Farm AI Deployment in Salmon, While Feed Optimization Dominates in Shrimp

Despite being a larger market by volume, AI tools are less developed and deployed in the shrimp sector than in salmon. Experts attribute this to the lower value of shrimp per tonne, and more fragmented ownership: unlike salmon, which is farmed predominantly by

well-capitalized companies (see also [The Fish Site, 2011](#)), shrimp production spans family-run smallholdings to corporate producers (see also [Shrimp Welfare Project, 2021](#)), limiting the commercial viability of AI adoption for a greater fraction of producers.

Expert 1 ranked biomass estimation highest among in-water tools for salmon. Expert 2 reported that bioscopes—tools that simultaneously capture biomass, lice counts, wounds, respiration, and swimming behavior—are widely deployed across Chile and Norway, with biomass estimation having gained the most traction among their functions. Biomass is the primary asset farmers are selling, and knowing its current value and trajectory through a production cycle allows farmers to monitor and protect profitability. Biomass data also informs slaughter scheduling: weaker groups can be harvested first to avoid mortality that would render fish unsaleable, while healthier groups remain in the sea to continue growing, maximizing yield and revenue. Biomass is furthermore a key input to calculating feed quantities. The experts we spoke with estimate that initial stocking numbers can be off by 5–20%, which can result in significant under- or overfeeding. Given that feed is the dominant production cost (see [Part 1](#)), biomass estimation is particularly economically valuable.

Feed optimization tools appear less embedded in salmon farming. Expert 1 ranked these last among AI product categories, attributing this to the technical complexity of modeling feeding behavior and integrating with large feeding systems. Expert 2 noted that uptake is stronger in Chile than Norway, where feed represents a larger share of margins and management culture favors automating feeding decisions.

In contrast, **Expert 3 reported that feed optimization tools dominate shrimp AI deployment.** Traditionally, feed quantities were calculated from population estimates—you needed to know how many animals you had to determine what percentage of their total body weight to feed. AI-managed feeding has largely bypassed this problem: rather than calculating feed from population, systems now use hydroacoustic sensors⁷ to detect whether animals are actively eating, and feed on that basis. The expert suggested that they would now expect further gains to come from broader adoption of existing technology, rather than improvements to current technologies. The expert noted that **without AI-enabled feeding technology, it is becoming increasingly difficult for shrimp producers to reach the cost levels required to remain competitive.**

Population counting and estimation tools were ranked second by Expert 3 for use in shrimp. These include:

⁷ Hydroacoustic sensors (commonly hydrophones, which function like underwater microphones) detect the clicking sounds shrimp make when eating, and use these to determine whether the animals are actively feeding, triggering feeders to dispense or withhold feed accordingly.

- **Larval stage:** where counting small animals can have a high error rate and can cause damage to the animals.
- **Grow-out stock surveys:** where shrimp are sampled from the pond and assessed using computer vision.
- **Post-harvest:** where computer vision sorting technologies are used in factory settings to separate shrimp from pond debris, such as stones and wood, before processing (though it was not established whether the shrimp would be alive or dead at this stage).

The Primary Barrier to AI Adoption is an Unclear Return on Investment

In addition to context from our expert interviews, the following section draws on a Claude-assisted review of 16 academic papers published between 2023 and 2026 (four primary empirical studies and 12 literature reviews), of which 13 focus primarily on aquaculture and three draw on broader agricultural or animal farming contexts. See the [Appendix](#) for more information.

Cost and unclear return on investment are the most consistently cited barriers to AI adoption in aquaculture across the papers that address this question directly ([Georgopoulos et al., 2023](#); [León-Balderrama et al., 2026](#); [Chowdhury et al., 2025](#); [Tina et al., 2025](#); [Areeba et al., 2025](#)). This is independently agreed by our experts, who suggest that tools with demonstrable, measurable returns—such as biomass estimation and feed optimization—have gained broader traction than those whose benefits are harder to quantify. Expert 2 added that the return on investment in technological upgrades is further reduced in countries where more pronounced seasonal climates can mean there are periods of the year when no farming takes place.

A further barrier is data availability, quality, and willingness to share. Our experts note that records are typically held in disparate formats, and that farmers are reluctant to share data they consider proprietary, limiting the volume of data that AI tool developers can train their models on. The literature points to two related problems:

First, **producing large, well-annotated datasets is costly** ([Wu et al., 2025](#)), **labor-intensive** ([Wu et al., 2025](#); [D'Agaro, 2025](#)), and the data is **technically challenging to collect** ([Wu et al., 2025](#); [D'Agaro, 2025](#); [Akram et al., 2026](#)). Annotated samples are also severely imbalanced across fish species and behavior categories ([Wu et al., 2025](#)). This can lead, as observed by Expert 1, to **tools that are not universally suitable across all farm types and species.**

Second, **proprietary data concerns can deter adoption altogether**, particularly where it is unclear who controls and benefits from farm data ([Manning, 2024](#); [Ryan & Blok, 2026](#)). Expert 3 noted that in one project where data was shared anonymously across multiple producers, model performance was substantially better than when trained on a single farm's data alone—suggesting data pooling may be important for model quality. No paper in our corpus directly tests this.

Expert 1 noted that a related challenge is product maturity: **AI companies often release products before they are sufficiently mature, requiring producers to be patient while tools develop**. One such example described was an automated feeding system that turned itself on unsupervised in the early hours and dispensed 11 tons of feed into a single cage. Some companies are attempting to address contextual variability through structured on-farm calibration periods—in one case requiring three full grow-out cycles of data collection before the tool is fully operational ([Fletcher, 2025](#))—though the extent to which this resolves the underlying data limitations is unclear.

Skill gaps are a consistently cited barrier to AI adoption, primarily reflecting low digital and AI literacy among farm operators and staff ([Georgopoulos et al., 2023](#); [Chowdhury et al., 2025](#); [Gkikas et al., 2024](#)). Expert 1 noted that training one shift of staff does not guarantee continuity, and untrained staff can inadvertently disable or misuse systems. However, one study of smallholder farmers found that self-assessed digital competence did not significantly predict AI adoption intention ([Cao et al., 2026](#)), suggesting that addressing skills may be necessary but not sufficient.

Government Action Contributes to Shaping AI Development and Adoption in Aquaculture

Several governments have established dedicated aquaculture R&D and innovation funding programs—among them Norway ([Norwegian Ministry of Trade, Industry and Fisheries, 2023](#)), Scotland ([Fish Farming Expert, 2026](#)), Canada ([Responsible Seafood Advocate, 2025](#)), and the EU ([European Commission, 2025](#))—which have funded or explicitly include AI and digital technology applications, though we have not conducted an in-depth search for funding sources or assessed the extent to which AI is prioritized across these programs.

Public investment can also be particularly important where the commercial case for AI-adoption is weak. Expert 1 reported that **the weaker return on investment case for lower-value species discourages both technology development by suppliers and adoption by farmers**. The expert suggested that China has lagged for this reason, though government investment in university research is reported to be accelerating the development of AI tools suited to China's specific species and production systems. In addition, effective May 2026, **China's revised Fisheries Law explicitly promotes a shift from extensive to intensive** and

environmentally friendly aquaculture production ([Godfrey, 2026](#)), **which may bring about more AI adoption.**

Regulatory frameworks can also act as a bottleneck to AI adoption. Norwegian rules currently require manual lice counting of at least 20 fish per site weekly, and farms wishing to use automated camera systems in place of manual counts must obtain individual government dispensation to do so. The Norwegian Food Safety Authority has granted a large number of exemptions and is actively preparing formal regulatory requirements for automated methods, with Standards Norway developing minimum technical specifications ([Ministry of Trade, Industry and Fisheries, 2025](#), p. 76). Expert 3 noted that mandatory automated lice counting is anticipated in the near term, which would accelerate uptake and eliminate the physical fish handling that manual counting requires.

Experts Predict That AI Tool Uptake Will Increase Rapidly in the Short-Term

The United Nations' Food and Agriculture Organization (FAO) projects global aquaculture production to grow over the coming decade ([FAO, 2024](#)). Within this expanding industry, **our experts anticipate AI-tool uptake to increase significantly in the next 2–5 years.** We expect this to be particularly true of tools with the clearest return on investment. Expert 3 noted that without AI-enabled feeding technology, it is becoming increasingly difficult for shrimp producers to keep feed costs low enough to remain competitive. The expert projected that AI could be deployed across more than 70% of farmed hectares in Ecuador within five years, attributed partly to the rapid consolidation of large producers acquiring smaller farms and rolling out technology at scale, recovering their investment quickly through the resulting profits.

With time, we expect AI tools to become cheaper and easier to extend to more species.

Training data requirements for feed optimization models are substantial: Expert 1 estimated that developing a reliable salmon feeding model required 14,000 cage days of data, though noted that applying an existing framework to a new species—such as sea bass or sea bream—may require only around a third of that. The reasons for this reduction were not established, though one possibility is transfer learning: a model trained on one species may encode general feeding-behavior patterns that require only limited additional data to adapt to a related species, rather than training from scratch. The literature cautions, however, that **models do not always transfer readily across species or farming contexts, owing to differences in behavioral traits and environmental conditions** ([Huang & Khabusi, 2025](#); [Tina et al., 2025](#); [D'Agaro, 2025](#)). Even where a business case exists, the pathway to deployment is likely to be longer and more resource-intensive for species outside the best-studied systems.

Conclusions

This second installment of How AI is Affecting Farmed Aquatic Animals maps where the AI-enabled tools identified in [Part 1](#) are being deployed in practice. We found that deployment is concentrated in Europe, Latin America, and Southeast Asia, with Norway and Scotland the largest exporters. In our dataset, Chile receives more deployments from foreign-headquartered companies than any other single country, while having no domestically-headquartered AI companies of its own. Deployment reaching lower- and middle-income producers across Latin America and Southeast Asia suggests that the welfare implications of AI in aquaculture could extend beyond the high-value, high-income operations that tend to dominate discussion of the sector.

Of the species groups in our [database](#), salmon and shrimp have the highest number of deployment instances in absolute terms, consistent with—though not independent of—their dominance at the innovation stage. Salmon AI deployment is more geographically concentrated than shrimp, particularly around Norway and Chile; shrimp AI deployment is more dispersed across Latin America and Southeast Asia. AI adoption currently remains limited: experts estimate approximately 15% of all salmon producers and <10% of shrimp producers use AI tools—a figure that rises to ~75% and ~25%, respectively, among top producers. The primary barrier to greater adoption is reported to be an unclear return on investment: tools with demonstrable returns, such as feed optimization and biomass estimation, have gained the broadest traction.

In our final report, we will examine what these patterns mean for the welfare of the hundreds of billions of aquatic animals farmed each year.

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If you are interested in RP's work, please visit our research database and subscribe to our newsletter.

Appendix

Definitions

To ensure that our deployment analysis reflects AI adoption in aquaculture specifically, we classified companies by the degree to which they specialize in aquaculture. We excluded multi-sector companies from the deployment analysis, as deployment locations recorded for these companies could reflect AI use in non-aquaculture contexts rather than in aquaculture. The definitions applied are as follows:

Box 3: Criteria Used to Designate Companies as Aquaculture-Only, Aquaculture-Primary, and Multi-sector

- **Aquaculture-only:** The company website references aquaculture or a specific farming type (e.g., salmon farming) with no mention of any other sector.
- **Aquaculture-primary:** The company predominantly operates in aquaculture but also has a presence in a closely related field, such as capture fisheries, fish tracking, population monitoring, water quality forecasting, or aquatic research. Of the five companies in this category:
 - Three companies have a presence in a related field, but aquaculture is the clear primary focus of their commercial positioning, and the related-field presence takes the form of on-request services, grant-funded research, or potential future applications.
 - Two companies apply their core technology in both aquaculture and fisheries.
- **Multi-sector:** The company operates across multiple sectors, of which aquaculture is not the clear primary focus, or has substantial operations in unrelated industries (e.g., oil and gas, general agriculture, food processing).

To assess where AI-aquaculture tools are being used in practice, we systematically searched for deployment evidence for each company in our database. The methods used and inclusion criteria applied are described below.

Box 4: Framework for Deducing the Countries to Which a Company Deployed its AI-Aquaculture Products

- We used Claude to conduct targeted web searches for company names + “deployment” + “locations” and related search terms (e.g., “farms,” “pilots”). We asked Claude to provide search results by deployment country, with links.
- Links provided by Claude were then checked manually.
- We recorded deployment locations where a company explicitly stated that it was operating or had deployed technology in a specific country, as found in company websites (including client directories and case studies), press releases, third-party articles, or other online resources.
- The method does not differentiate between a single deployment and multiple deployments within the same country.
- We included deployments at all stages—pilot, field trial, and commercial-scale.
- Deployment locations were recorded at the company level rather than the product level.
- We excluded countries mentioned only as target markets, future expansion plans, technology development partnerships, or office locations without corroborating evidence of on-site product use.
- Only companies primarily operating in aquaculture were included; companies with substantial operations in unrelated industries were excluded.
- Companies for which no deployment information could be found in publicly available sources were not assigned any deployment locations.

Expert Profiles

Table 2: Overview of the background and experience of the three experts interviewed for this report

	Expert 1	Expert 2	Expert 3
Role type	Professor of aquaculture & CEO of an aquaculture technology consultancy	Former Chief Technology Officer at an aquaculture AI company & independent consultant	Advisor to an aquaculture technology company & co-founder of an aquaculture consultancy
Species expertise	Salmonids, shrimp, bluefin tuna	Salmonids	Shrimp
Years in aquaculture	~30	~5 (+~10 in AI)	~12

Based in	Australia	Norway	Spain
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Claude-Review of Academic Papers on Barriers to Adoption of AI in Aquaculture

We completed a timeboxed three-hour search to find and screen papers discussing barriers to AI adoption in aquaculture. We used Claude to identify relevant papers, manually checked titles for relevance, and retained 16 papers for which the full text was available. Of these, 13 focused primarily on aquaculture while three drew on wider agricultural or animal farming contexts.

Of the 16 papers, 12 are literature reviews (four systematic reviews using documented search protocols, and eight narrative syntheses), and four are empirical studies collecting data directly from farmers, industry professionals, or stakeholders (from Greece, the Netherlands, Canada, and China). The depth of focus on adoption barriers varies considerably across the set, and studies examining adoption from the farmer or operator perspective remain sparse.

We asked Claude to extract the main barrier themes from each paper, noting whether a theme was addressed and, where it was, how prominently it featured as a discouraging factor (for example, a numerical ranking, a percentage share of cited barriers, or explicit characterization such as "the most commonly cited barrier"). We then compared these findings against the views expressed by our expert interviewees, and selected the themes with the strongest consensus or most substantive discussion across both sources for inclusion in this report.

It is important to note that the themes highlighted here may not fully represent the barriers faced by practitioners implementing AI in aquaculture. The papers in this selection vary considerably in their framing, objectives, and methodology. Of the 16 papers, only eight treat barriers as a central analytical focus, four address them as a secondary concern, and two mention them only incidentally.

Ways in Which We Could Be Wrong

The lists below are non-exhaustive.

How Our Database Could Be Incomplete or Inaccurate

We should indicate if and how this exclusion would most distort the picture we are painting (i.e., if AI products from multi-sector companies have a different pattern of deployment than their single-sector counterparts). We acknowledge potential biases from other limitations in data, but not this one.

Classification of aquaculture-only, aquaculture-primary, and multi-sector companies

- We acknowledge there is a degree of judgment in the classification between aquaculture-primary and multi-sector. For example, one company whose core business is aquaculture but which offers water quality forecasting services outside aquaculture on request was placed in the aquaculture-primary category, whereas a company that specialized in marine energy, reef monitoring, and academic research before moving into aquaculture was placed under multi-sector. The majority of cases are more clear-cut, and only five companies were placed under “aquaculture-primary.”
- The classification was applied at the company level, not the product level. A company classified as aquaculture-primary may have individual products deployed exclusively in aquaculture, or conversely, products deployed in non-aquaculture contexts.
- Where a company website made no mention of sectors other than aquaculture, we classified it as aquaculture-only. However, if such a company also deploys its technology in non-aquaculture contexts (e.g., oil and gas, environmental monitoring) without disclosing this publicly, we may have inadvertently recorded non-aquaculture deployments as aquaculture deployments.
- By restricting our analysis to aquaculture-only and aquaculture-primary companies, we may be misrepresenting regional deployment, deployment intensity, and/or the species affected if multi-sector companies deploy AI differently than aquaculture-specialist companies.

Logging of deployment locations

- Sources confirming a company is operating in a given location do not always confirm that the specific product deployed is AI-enabled. Where a company produces both AI-enabled and non-AI-enabled products, we may have recorded a deployment location where only the latter was in use.
- Sources describing deployments may be out of date—an article from 2022 reporting a deployment may not reflect whether that deployment is still active. Our database records locations as found at the time of research and does not capture discontinuations.
- Our database records whether a company has deployed in a country, not how extensively. A company with one installation in country X is counted the same as a company with 100 in the same country. When we refer to countries with the highest number of companies deploying, this should not be read as a measure of the volume or scale of deployment but of the number of distinct companies with a presence in that country.

- At the time of writing, some webpages identified during our earlier Innovation paper research were no longer accessible. This may indicate that some products have been discontinued or companies have ceased operating, in which case our database may overstate current deployment.
- There may be cases in which we are underestimating deployment: if a company stated it operated in a given number of countries without specifying which, we recorded only those for which we found supporting evidence.
- The sources of information used are biased towards English-language content. Companies operating primarily in non-English-speaking markets may be underrepresented if their deployment evidence exists only in local-language sources.
- Deployment locations were identified through targeted web searches rather than a systematic audit of every company's full client base. Deployments not publicly announced or documented online would not have been captured.

Limitations in what the database covers

- Companies may market products as AI-enabled when the AI component is minimal or peripheral. Where farms adopt such products, this may overstate the extent of genuine AI adoption in practice.
- Farms may develop in-house AI tools that are not reported in the public domain.

Barriers to adoption

The barriers to adoption identified in the literature and by experts may not capture the full picture. One mechanism that could further suppress adoption—but which we are not aware of being documented—concerns regulatory disincentives. While AI tools could help farms monitor fish health and minimize stock losses, certain voluntary regulatory frameworks may inadvertently discourage their adoption. For example, farmers might discontinue using AI systems that provide more accurate data if that transparency leads to adverse consequences, such as:

- Higher penalties resulting from precise sea lice counts that exceed regulatory thresholds
- Stocking restrictions triggered by accurate biomass measurements that reveal farms are approaching or exceeding permitted densities
- Regulatory scrutiny or intervention triggered by AI-detected welfare indicators or real-time mortality data

Regional Groupings

- **Africa:** Ghana, Kenya, Lesotho, Nigeria, Rwanda, Tunisia, Uganda, Zambia
- **East Asia:** China, Japan, South Korea, Taiwan
- **Europe:** Bulgaria, Croatia, Cyprus, Denmark, England, Faroe Islands, Finland, France, Germany, Greece, Iceland, Ireland, Italy, Lithuania, Moldova, Netherlands, Norway, Poland, Portugal, Romania, Russia, Scotland, Slovakia, Spain, Sweden, Switzerland
- **Latin America:** Argentina, Brazil, Chile, Colombia, Costa Rica, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Peru
- **Middle East:** Armenia, Georgia, Israel, Saudi Arabia, Turkey
- **North America:** Canada, US
- **Oceania:** Australia, New Zealand
- **South Asia:** Bangladesh, India
- **Southeast Asia:** Brunei, Cambodia, Indonesia, Malaysia, Myanmar, Philippines, Singapore, Thailand, Vietnam

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